Environment

Introducing Environmental Science and Sustainability

Raven

Food as a Lens for Our Relationship to the Environment

- Chicken sandwich requires: wheat, chicken, other ingredients, pesticides, fertilizers, energy (petroleum) to manufacture, transport, treat generated wastes, and make packaging, landfills
- Our individual choices affect the environment
- How could we adapt our food-production practices for greater sustainability?

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The Environment (Earth)

- Life has existed on Earth for 3.8 billion yrs.
- Earth well suited for life
  - Water over $\frac{3}{4}$ of planet
  - Habitable temperature, moderate sunlight
  - Atmosphere provides oxygen and carbon dioxide
  - Soil with essential minerals for plants
- Modern humans appeared in Africa only 100,000 years ago

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Increasing Human Numbers

- In 1950, eight cities had populations > 5 million
  - NYC -12.3 million
- In 2016, large urban agglomerations of cities
  - 10 largest metropolitan areas, total of 200 million
  - Tokyo, Japan – 17.8 million, with 38.1 in greater metropolitan area

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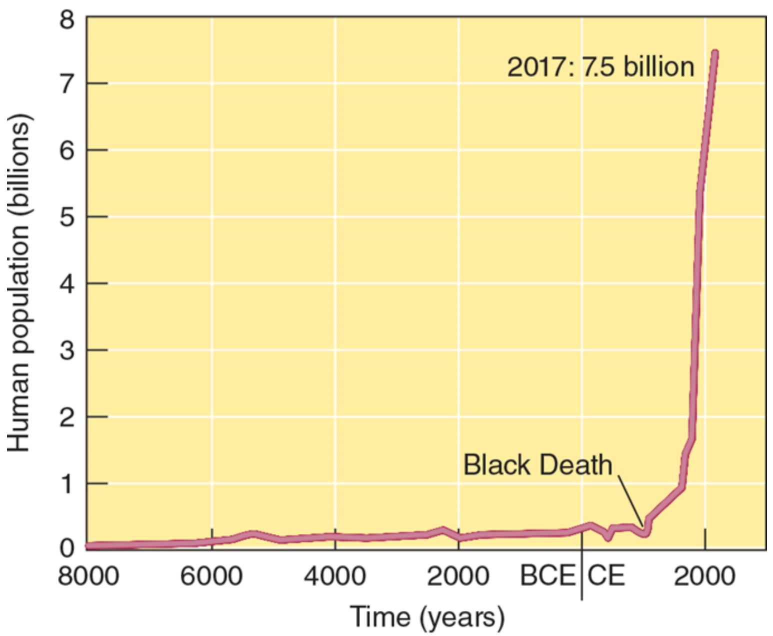
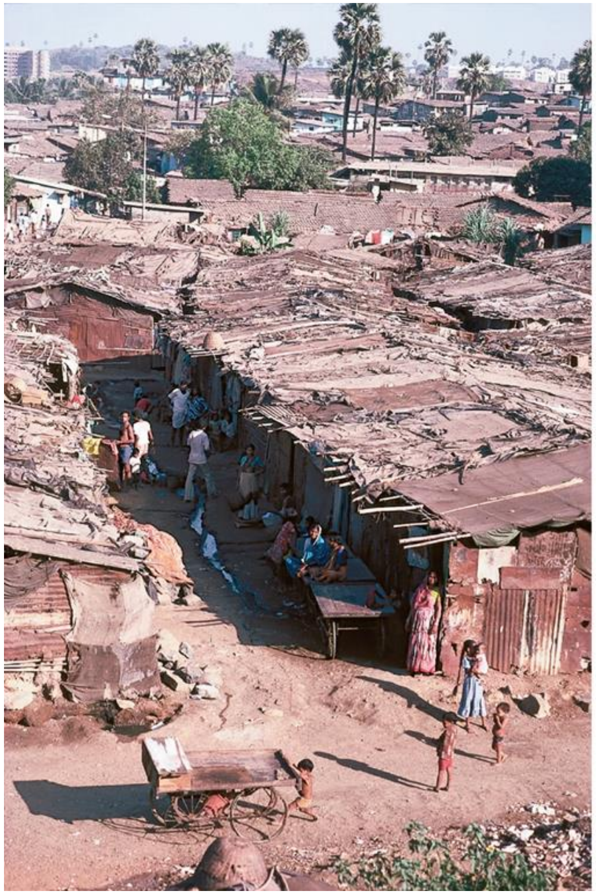
Human Population Growth

- In 2017, human population is : 7.5 billion
  - Passed 7 billion in 2011
  - Growing exponentially
  - Estimates of 9.3 -10.5 billion by end of 21st century
- In areas with fast growing populations, quality of life for many people may worsen considerably



Population and Extreme Poverty

- More than 1 in 2 people live in extreme poverty
  - Cannot meet basic needs for food, clothing, shelter, health
  - < $2.50 per U.S. dollars day
- Fertility rates worldwide ~ 3 children per family
  - Expected to decline and stabilize by end of 21st century
- Difficult to meet population needs without exploiting Earth's resources



Countries Differentiated Based on Wealth

- Highly Developed Countries (HDC)
  - Complex industrialized bases, low population growth, high per capita incomes
  - Ex: U.S., Canada, Japan
- Less Developed Countries (LDC)
  - Developing countries, low level of industrialization, high fertility rate, high infant mortality rate, low per capita income (relative to highly developed countries)
  - Ex: Bangladesh, Kenya, and Nicaragua

Income Disparity Between Rich and Poor

- Rising income disparity in many countries
  - Large gap between wealthy and poor citizens
    - Differential access to electricity, cars, modern medicine
  - Recently, greater disparity between urban and rural citizens in LDC
  - Ex: China, India, Brazil, Mexico
- Total wealth of country less accurate in describing well-being of citizens

Population, Resources, and Environment: Needs for Survival

- Two useful generalizations:
  - 1- Essential resources for individual survival are small
    - However, rapidly increasing population can quickly overwhelm or deplete, especially locally

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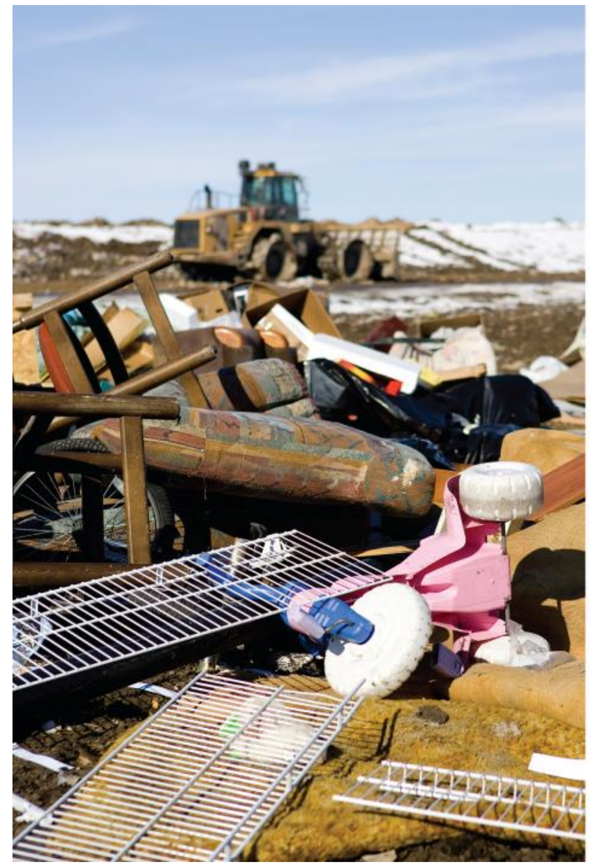
Individual Resource Consumption

- 2- Individual resource consumption can far outweigh needs of survival
  - Affluent nations use larger portions and can exhaust resources globally



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Types of Natural Resources

Natural Resources

- **Renewable Natural Resources**
  - Direct solar energy
  - Energy of winds, tides, flowing water
  - Fertile soil
  - Clean air
  - Fresh water
  - Biological diversity (forests, food crops, fishes)
- **Nonrenewable Natural Resources**
  - Metallic minerals (gold, tin)
  - Nonmetallic minerals (salt, phosphates, stone)
  - Fossil fuels (coal, oil, natural gas)

Resource Consumption

- Human use of materials and energy
  - Economic and social act
- People in HDCs are big consumers
- Unsustainable Consumption
  - When level of demand on resources damages or depletes resources enough to reduce the quality of life for future generations
  - Caused by overpopulation and/or overconsumption

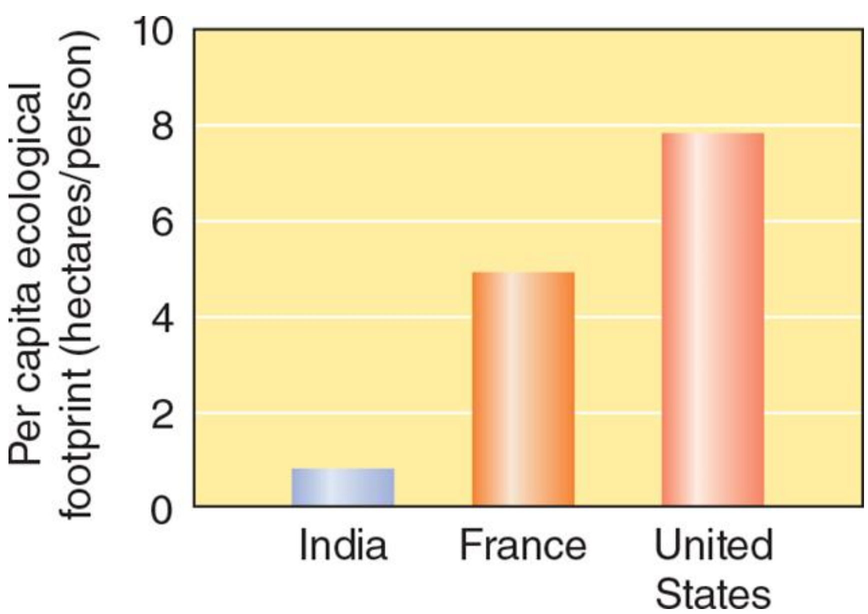
Ecological Footprint

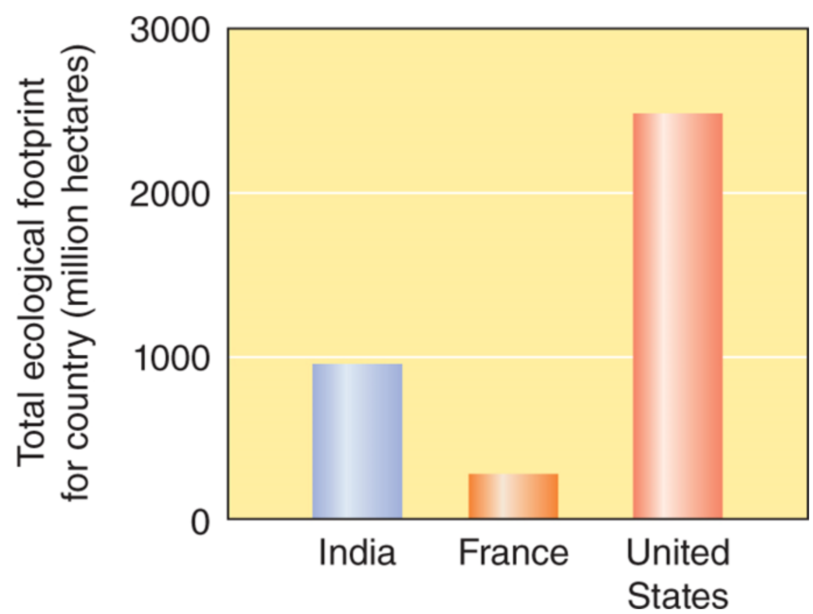
- The average amount of productive land, fresh water, and ocean required to continuously provide that person with all the resources they consume

Earth's Productive Land and Water	11.4 billion hectares
Amount Each Person is Allotted (divide Productive Land & Water by Human Pop.)	1.5 hectares
Current Average Global Ecological Footprint	2.7 hectares

Ecological Footprint Comparison

- Large variation in footprints among countries





Sustainability Requires Long-term Perspective (1 of 2)

- Ability to meet current human economic and social needs without compromising the ability of the environment to support future generations

Sustainability Requires Long-term Perspective (2 of 2)

Focus on Sustainability

- Stabilize human population
- Prevent pollution where possible
- Restore degraded environments
- Protect natural ecosystems
- Use resources efficiently
- Educate all boys and girls
- Prevent and reduce waste
- Eradicate hunger and poverty

Sustainability and Human Behavior

- Not often operating sustainably because of behaviors:
  - Extract resources as if they are unlimited
  - Consume faster than resources are replenished
  - Pollute at high rates
  - Increase population despite finite resources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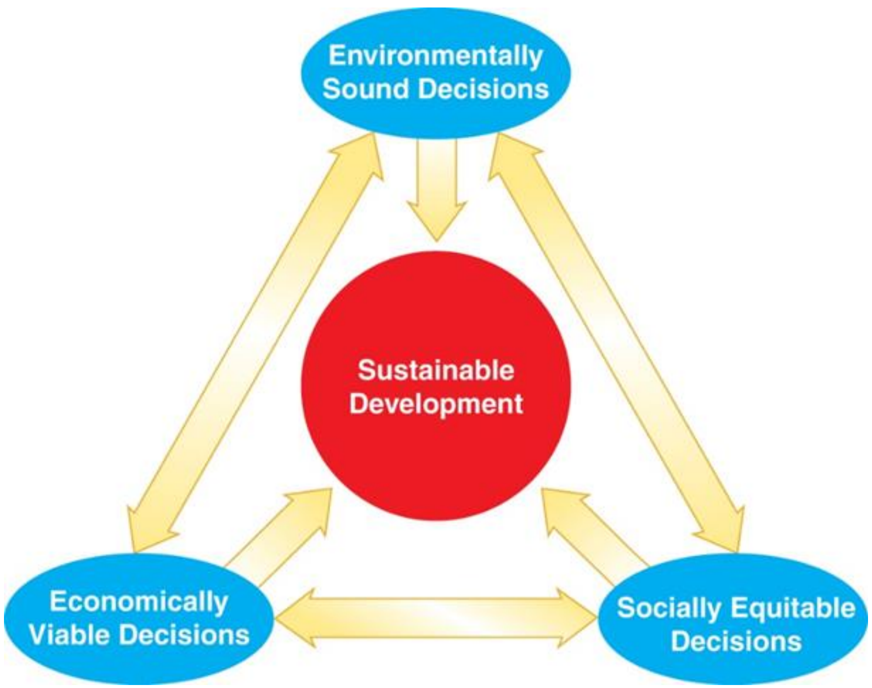
Sustainable Development- Systems Concept

- Economic development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs
- Summits helping to form international approaches



Environmental Science

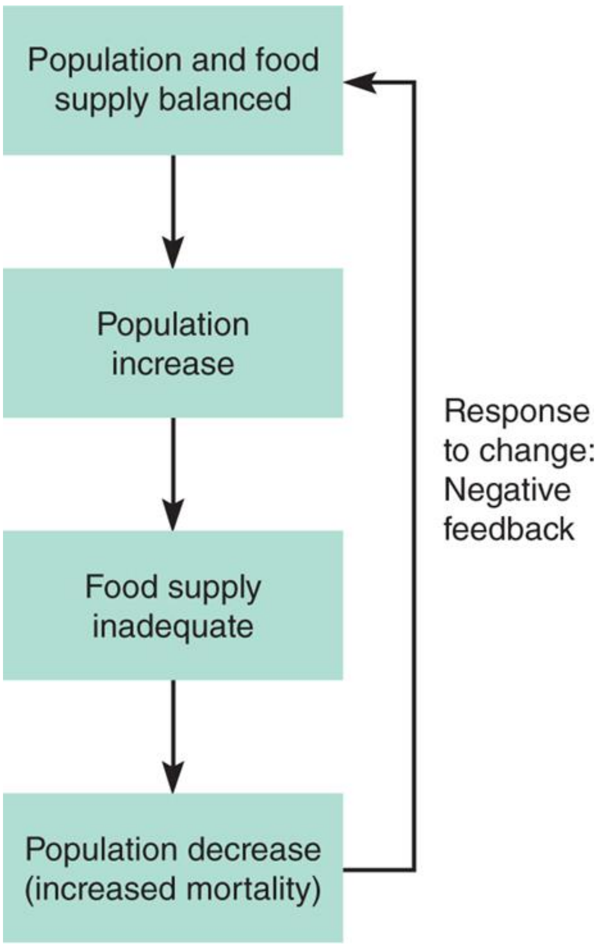
- Interdisciplinary study of humanity's relationship with other organisms and the nonliving physical environment
  - Biology
  - **Ecology**
  - Geography
  - Chemistry
  - Geology
  - Physics
  - Economics
  - Sociology
  - Demography
  - Politics

Earth's System and Environmental Science

- System
  - A set of components that interact and function as a whole
- Global Earth Systems
  - Climate, atmosphere, land, coastal zones, oceans
- Ecosystem
  - A natural system consisting of a community of organisms and its physical environment
- Systems in a dynamic equilibrium with feedback among components

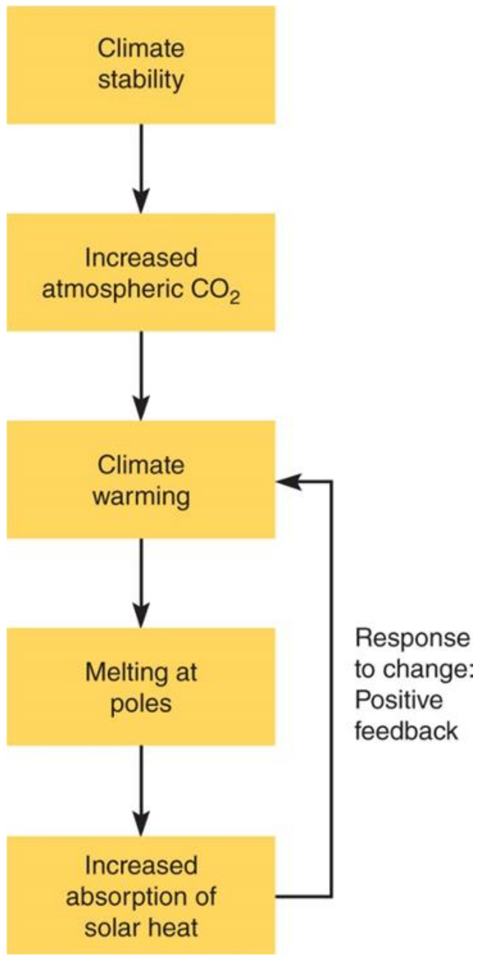
Negative Feedback

- Negative feedback
  - Change in some condition triggers a response that counteracts (reverses) the changed condition



Positive Feedback

- Positive feedback
  - Change triggers a response that intensifies the changing condition
  - Ex: polar and glacial ice melt, color change leading to more rapid melting



Climate Change: Hypotheses and Theory

- CO_2 and other gases from burning fossil fuels affect climate
- Unable to test or run large experiments globally
  - Must acquire lots of data and adapt theories and understanding with new data
- Many parts of climate theory are tested
  - CO_2 and rise in atmosphere
  - Impact of gases on solar radiation

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